

Rate-Distortion Guided Bottleneck Sizing for Quantized Autoencoders

TMLR-Oriented Draft, v0.2 with Mathematical Appendix

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May 2026

Abstract

Autoencoders often require a bottleneck dimension to be selected before deployment or large-scale model selection. In practice, this dimension is usually chosen by exhaustive sweeps over candidate sizes, which can be expensive when each candidate must be trained, quantized, and evaluated. We propose a rate-distortion guided sizing rule for quantized autoencoders. The method defines an operational latent rate under a fixed quantization, normalization, and factorized entropy coding protocol, and compares this rate to a target rate derived from rate-distortion theory. The ideal target is the data distribution’s rate-distortion function at a chosen saturation distortion, $R_X(D_{\text{sat}})$. Since $R_X(D)$ is generally unknown, experiments use an operational estimate obtained from a reference compressor, $\hat{R}_{\text{ref}}(D_{\text{sat}})$. The resulting dimension is the smallest candidate whose measured latent rate reaches the reference rate. We provide a finite-alphabet formulation, clarify the relationship between factorized latent rate and joint entropy through total correlation, derive a rate-distortion necessary condition, and give a condition under which the predicted crossing corresponds to bounded marginal reconstruction improvement. The proposed framework is intended as a design-time predictor of the quality-saturation region, not as a lossless minimum-dimension theorem.

1 Introduction

Autoencoders (AEs) are widely used as compact representations for reconstruction, compression, denoising, anomaly detection, and downstream inference. A central design choice is the bottleneck dimension d . A dimension that is too small can degrade reconstruction quality, while a dimension that is too large increases computation, memory, and encoded payload size. The common practice is to train or profile many candidate dimensions and select the smallest one that satisfies a quality constraint. This approach is simple but inefficient, especially when candidate dimensions are numerous or when quantized deployment must be evaluated separately for each candidate.

This draft develops a rate-distortion guided rule for selecting the bottleneck dimension of quantized autoencoders. The proposed rule is based on two operational quantities. First, for a fixed quantization and normalization protocol, we measure the factorized latent coding rate

$$R_{\text{fac}}(d) = \sum_{j=1}^d H(\hat{Z}_j), \quad (1)$$

where $\hat{Z}_j$ is the quantized j -th latent coordinate. Second, we define the target rate through rate-distortion theory. The ideal target is

$$H_{\text{target}}^* = R_X(D_{\text{sat}}), \quad (2)$$

where $R_X(D)$ is the rate-distortion function of the data distribution and D_{sat} is a user-specified saturation distortion. Since $R_X(D)$ is generally not available in experiments, we use an operational reference rate

$$H_{\text{target}} = \hat{R}_{\text{ref}}(D_{\text{sat}}), \quad (3)$$

measured from a reference compressor at the same distortion level.

The predicted saturation dimension is then

$$d_{\text{sat}} = \min \left\{ d \in \mathcal{D} : R_{\text{fac}}(d) \geq \hat{R}_{\text{ref}}(D_{\text{sat}}) \right\}, \quad (4)$$

where $\mathcal{D}$ is the set of candidate bottleneck dimensions. This rule is not intended to prove that d_{sat} is the minimum dimension required for reconstruction. Rather, it predicts the onset of a region in which additional latent dimensions provide diminishing reconstruction-quality improvements.

Contributions. This draft makes the following contributions.

1. It formulates AE bottleneck sizing using discrete quantized latent variables and an operational factorized coding rate.
2. It replaces input-entropy proxy targets with a rate-distortion target: the ideal target is $R_X(D_{\text{sat}})$, and the practical target is the measured reference compressor rate $\hat{R}_{\text{ref}}(D_{\text{sat}})$.
3. It derives a necessary condition connecting achievable distortion and latent rate: any quantized AE that achieves distortion at most D must have a factorized latent rate at least as large as $R_X(D)$.
4. It gives a saturation interpretation under a local flatness assumption on the rate-distortion curve, yielding a bound on marginal improvement after the target-rate crossing.
5. It proposes a TMLR-oriented experimental protocol for testing prediction accuracy, search-cost reduction, and robustness across datasets and quantization protocols.

2 Related Work

Rate-distortion theory. Rate-distortion theory formalizes the minimum mutual information required to reconstruct a source within a prescribed distortion level. It provides the natural target for lossy reconstruction problems, in contrast to lossless entropy. Our formulation uses $R_X(D)$ as the ideal target for AE bottleneck sizing.

Learned compression. Learned image compression methods optimize neural encoders, decoders, and entropy models under rate-distortion objectives. These methods typically learn an entropy model and directly optimize bitrate and distortion. Our goal is different: we do not propose a new compressor, but use a reference rate-distortion point to select the bottleneck dimension of a quantized AE.

Information bottleneck and representation learning. Information bottleneck methods analyze representations through mutual information and compression. VAEs and related latent-variable models also use information-theoretic quantities such as KL divergence. In contrast, the proposed method focuses on a fixed quantization protocol and the operational rate of a finite-alphabet latent representation.

Bottleneck dimension selection. Prior approaches often select AE bottleneck sizes empirically, by reconstruction-quality sweeps, validation curves, or heuristic rules. The proposed method aims to reduce this search by predicting the saturation region from a measured latent coding rate and a reference rate-distortion target.

3 Problem Setup

Let $X \sim P_X$ be a data sample in $\mathcal{X}$. A deterministic encoder and decoder are defined as

$$f_\theta : \mathcal{X} \rightarrow \mathbb{R}^d, \quad g_\phi : \mathcal{K}^d \rightarrow \hat{\mathcal{X}}, \quad (5)$$

where $\mathcal{K}^d$ denotes a finite quantized latent alphabet. The continuous latent vector is

$$Z = f_\theta(X) \in \mathbb{R}^d. \quad (6)$$

Before quantization, a fixed normalization protocol T_d is applied:

$$U = T_d(Z). \quad (7)$$

For example, T_d may be defined by statistics estimated on the training set and then frozen for validation and testing. The quantized latent is

$$\hat{Z} = Q_{\Delta, B}(U) = (\hat{Z}_1, \dots, \hat{Z}_d), \quad (8)$$

where each coordinate has bin width Δ_j , clipping range $[-B_j, B_j]$, and finite alphabet $\mathcal{K}_j$.

The reconstruction is

$$\hat{X} = g_\phi(\hat{Z}). \quad (9)$$

Given a distortion function $\ell : \mathcal{X} \times \hat{\mathcal{X}} \rightarrow \mathbb{R}_{\geq 0}$, the reconstruction distortion is

$$D_{\text{AE}}(d) = \mathbb{E}[\ell(X, \hat{X})]. \quad (10)$$

The objective is to select a bottleneck dimension d that lies near the quality-saturation region while avoiding unnecessary dimension sweeps.

4 Operational Latent Rate

Definition 1 (Quantized latent entropy). *For coordinate j , the quantized latent entropy is*

$$H(\hat{Z}_j) = - \sum_{k \in \mathcal{K}_j} p_{j,k} \log_2 p_{j,k}, \quad p_{j,k} = \mathbb{P}(\hat{Z}_j = k). \quad (11)$$

Definition 2 (Factorized operational latent rate). *Under a factorized entropy coding protocol, the operational latent rate is*

$$R_{\text{fac}}(d; \theta) = \sum_{j=1}^d H(\hat{Z}_j). \quad (12)$$

This is the ideal expected code length of a factorized entropy coder that encodes coordinates independently.

This quantity is operational because it refers to the expected number of bits under a specified coding protocol. It is not the differential entropy of the continuous latent vector. Since $\widehat{Z}$ is finite-valued, all entropies used in the method are discrete Shannon entropies.

Definition 3 (Joint latent rate and total correlation). *The joint entropy of the quantized latent is*

$$R_{\text{joint}}(d; \theta) = H(\widehat{Z}_1, \dots, \widehat{Z}_d). \quad (13)$$

The total correlation is

$$\text{TC}(\widehat{Z}) = \sum_{j=1}^d H(\widehat{Z}_j) - H(\widehat{Z}_1, \dots, \widehat{Z}_d). \quad (14)$$

Proposition 1 (Factorized rate decomposition). *For any discrete latent vector $\widehat{Z}$,*

$$R_{\text{fac}}(d; \theta) = H(\widehat{Z}) + \text{TC}(\widehat{Z}), \quad (15)$$

where $\text{TC}(\widehat{Z}) \geq 0$. Therefore,

$$H(\widehat{Z}) \leq R_{\text{fac}}(d; \theta). \quad (16)$$

Equality holds if and only if the latent coordinates are mutually independent.

Proof. The identity follows from the definition of total correlation. Non-negativity follows from the subadditivity of entropy. Equality holds precisely when the joint distribution factorizes across coordinates. $\square$

Effective-rate correction. When dependence between latent coordinates is substantial, $R_{\text{fac}}(d)$ can overestimate the joint latent entropy. If an estimate or upper bound τ_d on total correlation is available, a conservative effective-rate estimate can be defined as

$$R_{\text{eff}}(d) = R_{\text{fac}}(d) - \tau_d. \quad (17)$$

The corresponding conservative rule is

$$d_{\text{sat}}^{\text{eff}} = \min \left\{ d \in \mathcal{D} : R_{\text{eff}}(d) \geq \widehat{R}_{\text{ref}}(D_{\text{sat}}) \right\}. \quad (18)$$

This correction is optional in the base method but useful in ablations.

5 Rate-Distortion Target

Definition 4 (Rate-distortion function). *For a distortion function ℓ , the rate-distortion function of X is*

$$R_X(D) = \inf_{p(\tilde{x}|x): \mathbb{E}[\ell(X, \tilde{X})] \leq D} I(X; \tilde{X}). \quad (19)$$

The ideal target rate for bottleneck sizing at saturation distortion D_{sat} is

$$H_{\text{target}}^* = R_X(D_{\text{sat}}). \quad (20)$$

This target is theoretically appropriate for lossy reconstruction because it measures the minimum information required to achieve distortion at most D_{sat} .

In practice, $R_X(D)$ is usually unknown. We therefore estimate the target with a reference compressor. Let c_m be a reference codec or learned compressor configured to operate near distortion D . For a validation set $\{x_i\}_{i=1}^n$, define

$$\widehat{D}_{\text{ref}}(m) = \frac{1}{n} \sum_{i=1}^n \ell(x_i, \widehat{x}_i^{(m)}), \quad (21)$$

$$\widehat{R}_{\text{ref}}(m) = \frac{1}{n} \sum_{i=1}^n \text{bits}_m(x_i). \quad (22)$$

The operational target is selected as

$$\widehat{R}_{\text{ref}}(D_{\text{sat}}) = \widehat{R}_{\text{ref}}(m^*), \quad m^* \in \arg \min_m \left| \widehat{D}_{\text{ref}}(m) - D_{\text{sat}} \right|. \quad (23)$$

If multiple reference compressors are available, a stronger operational target can be defined as

$$\widehat{R}_{\text{ref}}^{\min}(D_{\text{sat}}) = \min_{m \in \mathcal{M}: \widehat{D}_{\text{ref}}(m) \leq D_{\text{sat}}} \widehat{R}_{\text{ref}}(m), \quad (24)$$

which reduces the effect of an inefficient single reference codec.

6 Bottleneck Sizing Rule

Given candidate dimensions $\mathcal{D}$, a fixed quantizer $Q_{\Delta, B}$, a frozen normalization protocol T_d , and a reference target $\widehat{R}_{\text{ref}}(D_{\text{sat}})$, the proposed sizing rule is

$$d_{\text{sat}} = \min \left\{ d \in \mathcal{D} : R_{\text{fac}}(d) \geq \widehat{R}_{\text{ref}}(D_{\text{sat}}) \right\}. \quad (25)$$

If total-correlation correction is used,

$$d_{\text{sat}}^{\text{eff}} = \min \left\{ d \in \mathcal{D} : R_{\text{fac}}(d) - \tau_d \geq \widehat{R}_{\text{ref}}(D_{\text{sat}}) \right\}. \quad (26)$$

Nested latent option. For stable monotonicity, one may train a maximum-width latent model and evaluate prefixes

$$\widehat{Z}^{(d)} = (\widehat{Z}_1, \dots, \widehat{Z}_d). \quad (27)$$

Then

$$R_{\text{fac}}(d+1) = R_{\text{fac}}(d) + H(\widehat{Z}_{d+1}) \geq R_{\text{fac}}(d), \quad (28)$$

so the crossing rule is monotone by construction. If independently trained models are used instead, a monotone envelope can be applied:

$$\overline{R}_{\text{fac}}(d) = \max_{d' \leq d} \widehat{R}_{\text{fac}}(d'). \quad (29)$$

Algorithm. The design-time procedure is as follows.

1. Choose a target distortion D_{sat} and evaluate one or more reference compressors on the validation set.
2. Set $H_{\text{target}} = \widehat{R}_{\text{ref}}(D_{\text{sat}})$, or $\widehat{R}_{\text{ref}}^{\min}(D_{\text{sat}})$ when multiple references are used.
3. For each candidate AE dimension $d \in \mathcal{D}$, train or partially train a quantized AE under the fixed quantization and normalization protocol.

4. Estimate $\widehat{R}_{\text{fac}}(d) = \sum_{j=1}^d \widehat{H}(\widehat{Z}_j)$ from validation latents.
5. Select the smallest candidate dimension whose latent rate reaches the reference target.
6. Optionally profile only a local neighborhood around the predicted dimension to select the final model under additional quality or deployment constraints.

7 Theoretical Analysis

7.1 Necessary Condition from Rate-Distortion Theory

Theorem 1 (Latent rate necessary condition). *Consider a quantized AE with Markov structure*

$$X \rightarrow \widehat{Z} \rightarrow \widehat{X}. \quad (30)$$

If the AE achieves distortion at most D , i.e.

$$\mathbb{E}[\ell(X, \widehat{X})] \leq D, \quad (31)$$

then

$$R_X(D) \leq I(X; \widehat{X}) \leq I(X; \widehat{Z}) \leq H(\widehat{Z}) \leq R_{\text{fac}}(d). \quad (32)$$

Consequently,

$$\mathbb{E}[\ell(X, \widehat{X})] \leq D \implies R_{\text{fac}}(d) \geq R_X(D). \quad (33)$$

Proof. The inequality $R_X(D) \leq I(X; \widehat{X})$ follows from the definition of the rate-distortion function because $\widehat{X}$ is a feasible reconstruction at distortion at most D . The data processing inequality for $X \rightarrow \widehat{Z} \rightarrow \widehat{X}$ gives $I(X; \widehat{X}) \leq I(X; \widehat{Z})$. Since $\widehat{Z}$ is discrete, $I(X; \widehat{Z}) \leq H(\widehat{Z})$. Finally, $H(\widehat{Z}) \leq \sum_j H(\widehat{Z}_j) = R_{\text{fac}}(d)$ by subadditivity. $\square$

The theorem gives a one-sided statement. Reaching a rate target is necessary for achieving a given distortion, but it is not sufficient. Sufficiency would require assumptions on model class, training optimality, quantization, and representation efficiency. The proposed rule is therefore a predictor of the saturation region rather than a universal guarantee of reconstruction quality.

7.2 Saturation Interpretation

Let the distortion-rate function be

$$D_X(R) = \inf_{p(\tilde{x}|x): I(X; \tilde{X}) \leq R} \mathbb{E}[\ell(X, \tilde{X})]. \quad (34)$$

The saturation interpretation becomes meaningful when the rate-distortion curve is locally flat beyond a target rate.

Assumption 1 (Local rate-distortion flatness). *There exist R_{sat} and $\varepsilon \geq 0$ such that for all $R \geq R_{\text{sat}}$,*

$$0 \leq -D'_X(R) \leq \varepsilon. \quad (35)$$

Equivalently, for $R_2 \geq R_1 \geq R_{\text{sat}}$,

$$D_X(R_1) - D_X(R_2) \leq \varepsilon(R_2 - R_1). \quad (36)$$

Assumption 2 (Near-optimality at the first crossing). *At a dimension d_1 , the AE distortion is within γ_{d_1} of the rate-distortion optimum at its measured rate:*

$$D_{\text{AE}}(d_1) \leq D_X(R_{\text{fac}}(d_1)) + \gamma_{d_1}. \quad (37)$$

Theorem 2 (Bounded marginal improvement after saturation). *Let $d_2 > d_1$ and suppose*

$$R_{\text{fac}}(d_2) \geq R_{\text{fac}}(d_1) \geq R_{\text{sat}}. \quad (38)$$

Under local rate-distortion flatness and near-optimality at d_1 ,

$$D_{\text{AE}}(d_1) - D_{\text{AE}}(d_2) \leq \varepsilon [R_{\text{fac}}(d_2) - R_{\text{fac}}(d_1)] + \gamma_{d_1}. \quad (39)$$

Proof. By the definition of $D_X(R)$, any reconstruction whose mutual information is at most $R_{\text{fac}}(d_2)$ has distortion at least $D_X(R_{\text{fac}}(d_2))$. Thus

$$D_{\text{AE}}(d_2) \geq D_X(R_{\text{fac}}(d_2)). \quad (40)$$

Using near-optimality at d_1 ,

$$D_{\text{AE}}(d_1) - D_{\text{AE}}(d_2) \leq D_X(R_{\text{fac}}(d_1)) + \gamma_{d_1} - D_X(R_{\text{fac}}(d_2)) \quad (41)$$

$$\leq \varepsilon [R_{\text{fac}}(d_2) - R_{\text{fac}}(d_1)] + \gamma_{d_1}, \quad (42)$$

where the last step uses the local flatness assumption. $\square$

This theorem formalizes the intended interpretation of the crossing rule. If the reference target corresponds to a locally flat region of the rate-distortion curve, then additional rate beyond the target can produce only bounded marginal distortion improvement, up to the training gap at the first crossing.

7.3 Finite-Sample Entropy Estimation

In experiments, $H(\widehat{Z}_j)$ is estimated from validation samples. Let $m_j = |\mathcal{K}_j|$, let p_j be the true distribution of $\widehat{Z}_j$, and let $\widehat{p}_j$ be the empirical distribution. If

$$\|\widehat{p}_j - p_j\|_1 \leq \eta_j < 1, \quad (43)$$

then a finite-alphabet entropy continuity bound gives

$$|H(\widehat{p}_j) - H(p_j)| \leq \eta_j \log_2(m_j - 1) + h_2(\eta_j), \quad (44)$$

where h_2 is the binary entropy function. Therefore,

$$|\widehat{R_{\text{fac}}}(d) - R_{\text{fac}}(d)| \leq B_d(n, \delta) := \sum_{j=1}^d [\eta_j \log_2(m_j - 1) + h_2(\eta_j)] \quad (45)$$

with probability at least $1 - \delta$, whenever the coordinate-wise distribution estimates satisfy their corresponding ℓ_1 bounds.

A confidence-aware crossing rule is

$$d_{\text{sat}}^\delta = \min \left\{ d \in \mathcal{D} : \widehat{R_{\text{fac}}}(d) - B_d(n, \delta) \geq \widehat{R_{\text{ref}}}(D_{\text{sat}}) \right\}. \quad (46)$$

This rule is more conservative than the plug-in crossing because it requires the lower confidence bound of the latent rate to exceed the reference target.

8 Experimental Protocol for TMLR Submission

The empirical section should test whether the proposed rate-distortion guided crossing predicts the quality-saturation region and reduces model-selection cost. The experiments should avoid relying on a single dataset or a single entropy proxy.

8.1 Datasets and Tasks

A TMLR-oriented evaluation should include multiple domains:

- image reconstruction: MNIST, Fashion-MNIST, CIFAR-10, and optionally CelebA or TinyImageNet;
- sensor or time-series reconstruction: HAR, ECG, or industrial sensor data;
- anomaly detection: reconstruction-based anomaly datasets where bottleneck dimension affects both quality and detection score.

8.2 Reference Rate-Distortion Targets

For each dataset and distortion metric, build an operational reference curve:

$$\{(\hat{R}_{\text{ref}}(m), \hat{D}_{\text{ref}}(m)) : m \in \mathcal{M}\}. \quad (47)$$

Possible references include standard codecs for image data, learned compressors when available, or domain-specific compressors for time-series data. The target D_{sat} should be chosen from the flat region of the reference rate-distortion curve or from an application-level quality threshold. The target rate is then $\hat{R}_{\text{ref}}(D_{\text{sat}})$.

8.3 Evaluation Metrics

The core metrics are:

- predicted saturation dimension d_{sat} ;
- reconstruction quality: MSE, PSNR, SSIM, or task-specific distortion;
- marginal gain after the crossing: $D_{\text{AE}}(d) - D_{\text{AE}}(d')$ for $d' > d$;
- search-cost reduction relative to exhaustive dimension sweeps;
- robustness across quantization bin widths, clipping ranges, and normalization protocols;
- total-correlation sensitivity and the effect of optional effective-rate correction.

8.4 Baselines

The method should be compared with:

- exhaustive dimension sweep;
- fixed dimension heuristics;
- quality-only early stopping over candidate dimensions;
- random or grid search over dimensions;
- Bayesian optimization over bottleneck dimension;
- latent variance or PCA-based sizing heuristics;
- reference-rate target without measured latent-rate crossing.

8.5 Expected Tables and Figures

Table 1 gives a recommended reporting format.

Table 1: Recommended reporting template for each dataset.

Dataset	D_{sat}	$\hat{R}_{\text{ref}}(D_{\text{sat}})$	d_{sat}	best d	quality gap	search reduction
Dataset A	–	–	–	–	–	–
Dataset B	–	–	–	–	–	–
Dataset C	–	–	–	–	–	–

The main figures should show:

1. reconstruction distortion versus dimension d ;
2. latent rate $R_{\text{fac}}(d)$ versus reference target $\hat{R}_{\text{ref}}(D_{\text{sat}})$;
3. marginal quality gain after the crossing;
4. search cost versus quality gap for the proposed method and baselines;
5. sensitivity to quantization and total-correlation correction.

9 Discussion

What the rule means. The proposed dimension is a design-time estimate of the saturation region. It is defined relative to a fixed quantization, normalization, factorized coding protocol, distortion metric, and reference rate-distortion target. It should be interpreted as an operational sizing rule, not as a universal intrinsic dimension of the dataset.

Why the reference target matters. Using $\hat{R}_{\text{ref}}(D_{\text{sat}})$ aligns the AE bottleneck target with lossy reconstruction. This is stronger than targets based on lossless input entropy because the bottleneck is evaluated at a specific distortion level.

When the method can fail. The method can be inaccurate when the reference compressor is poorly matched to the data or distortion metric, when the latent coordinates are highly dependent and no correction is applied, when the quantization protocol is unstable across dimensions, or when AE training is far from the rate-distortion frontier.

How to strengthen the method. The most direct improvements are to use multiple reference compressors, apply total-correlation correction, train nested latent models to ensure monotone rates, and report confidence-aware crossings using finite-sample entropy bounds.

10 Limitations

The target $\hat{R}_{\text{ref}}(D_{\text{sat}})$ is an operational estimate, not the true $R_X(D_{\text{sat}})$. A weak reference compressor can overestimate the target rate and produce conservative dimensions. The factorized latent rate can exceed the joint latent entropy when coordinates are dependent. Entropy estimates require sufficient validation samples and can be biased for large alphabets. Finally, rate crossing is a necessary-condition-inspired predictor; it does not guarantee that the selected AE achieves the target distortion without adequate model capacity and optimization.

11 Conclusion

This draft proposes a rate-distortion guided method for sizing quantized autoencoder bottlenecks. The method defines a discrete operational latent rate under a fixed quantization and factorized coding protocol, replaces proxy input entropy targets with an ideal rate-distortion target $R_X(D_{\text{sat}})$, and uses a reference compressor rate $\hat{R}_{\text{ref}}(D_{\text{sat}})$ for practical experiments. The resulting crossing rule predicts a saturation dimension rather than a lossless minimum dimension. Theoretical analysis provides a rate-distortion necessary condition and a bounded-improvement interpretation under local flatness of the distortion-rate curve. The next step is a multi-dataset empirical evaluation showing that the predicted crossing reduces dimension-search cost while preserving reconstruction quality.

A Detailed Explanation of the Mathematical Formulation

This appendix gives an expanded explanation of the main mathematical objects used in the paper. The goal is to make the rate-distortion guided sizing rule interpretable without changing the main claim of the paper. The selected dimension is a design-time predictor of a saturation region, not a theorem about a universally minimal autoencoder dimension.

A.1 Autoencoder mapping and bottleneck dimension

The encoder and decoder are written as

$$f_\theta : \mathcal{X} \rightarrow \mathbb{R}^d, \quad g_\phi : \mathcal{K}^d \rightarrow \hat{\mathcal{X}}. \quad (48)$$

The encoder maps an input sample X to a d -dimensional latent vector,

$$Z = f_\theta(X) = (Z_1, \dots, Z_d). \quad (49)$$

The number d is the bottleneck dimension. The purpose of the proposed method is to select a dimension that is large enough to reach the reconstruction-quality saturation region, while avoiding unnecessary latent coordinates that increase model-selection cost and deployed payload.

A.2 Normalization and quantized latent variable

The quantized latent variable is defined as

$$\hat{Z} = Q_{\Delta, B}(T_d(f_\theta(X))). \quad (50)$$

This expression has three steps. First, the encoder produces the continuous latent vector $f_\theta(X)$. Second, a fixed normalization protocol T_d is applied. For example, T_d may use mean and variance estimated from the training set and then frozen. Third, a fixed quantizer $Q_{\Delta, B}$ maps the normalized latent vector to finite bin indices.

The quantizer has bin widths Δ_j and clipping bounds B_j . After quantization,

$$\hat{Z}_j \in \mathcal{K}_j, \quad (51)$$

where $\mathcal{K}_j$ is a finite alphabet. This finite-alphabet construction is essential: the paper uses discrete Shannon entropy, not differential entropy of a continuous latent vector. The resulting rate quantities are therefore interpretable as bit budgets under a specified coding protocol.

A.3 Coordinate-wise entropy

For the j -th quantized latent coordinate, the entropy is

$$H(\hat{Z}_j) = - \sum_{k \in \mathcal{K}_j} p_{j,k} \log_2 p_{j,k}, \quad p_{j,k} = \Pr(\hat{Z}_j = k). \quad (52)$$

This is the average number of bits required to encode coordinate j with an ideal entropy code matched to its marginal distribution. If $\hat{Z}_j$ almost always falls into the same bin, then $H(\hat{Z}_j)$ is small. If $\hat{Z}_j$ is spread across many bins, then $H(\hat{Z}_j)$ is larger.

This quantity depends on the fixed quantization and normalization protocol. Changing bin width, clipping range, or normalization changes the empirical distribution of $\hat{Z}_j$ and therefore changes its entropy. For this reason, the protocol must be fixed before comparing dimensions.

A.4 Factorized operational latent rate

The factorized operational latent rate is

$$R_{\text{fac}}(d) = \sum_{j=1}^d H(\hat{Z}_j). \quad (53)$$

It is the ideal expected code length of a factorized entropy coding protocol that encodes the quantized coordinates independently. This is an operational coding quantity: it describes the number of bits required by a specified factorized code, rather than claiming to be the exact information content of the full latent vector.

This interpretation is important. The proposed method does not require the coordinates to be truly independent. Instead, $R_{\text{fac}}(d)$ is the cost of a factorized coding scheme. If a joint entropy coder were available and accurately modeled the dependence among coordinates, it could require fewer bits.

A.5 Joint entropy and total correlation

The true joint entropy of the quantized latent vector is

$$H(\hat{Z}) = H(\hat{Z}_1, \dots, \hat{Z}_d). \quad (54)$$

In general,

$$H(\hat{Z}) \leq \sum_{j=1}^d H(\hat{Z}_j) = R_{\text{fac}}(d). \quad (55)$$

The gap is the total correlation,

$$\text{TC}(\hat{Z}) = \sum_{j=1}^d H(\hat{Z}_j) - H(\hat{Z}). \quad (56)$$

Therefore,

$$R_{\text{fac}}(d) = H(\hat{Z}) + \text{TC}(\hat{Z}). \quad (57)$$

If the latent coordinates are mutually independent, then $\text{TC}(\hat{Z}) = 0$ and the factorized rate equals the joint entropy. If the coordinates are dependent, then $\text{TC}(\hat{Z}) > 0$ and the factorized rate is larger than the joint entropy.

This relation clarifies the role of the independence assumption. The paper does not need to assert that the coordinates are perfectly independent. It is enough to state that $R_{\text{fac}}(d)$ is the rate of a factorized coding protocol. Optional total-correlation correction can be used when dependence is substantial.

A.6 Rate-distortion function

The rate-distortion function is

$$R_X(D) = \inf_{p(\tilde{x}|x): \mathbb{E}[\ell(X, \tilde{X})] \leq D} I(X; \tilde{X}). \quad (58)$$

The variable $\tilde{X}$ represents a reconstruction of X , and $\ell(X, \tilde{X})$ is the distortion measure. The constraint

$$\mathbb{E}[\ell(X, \tilde{X})] \leq D \quad (59)$$

requires the reconstruction to have average distortion no larger than D . Among all reconstruction channels satisfying that constraint, $R_X(D)$ is the minimum mutual information needed between the source and the reconstruction.

This is the correct ideal target for lossy reconstruction. A lossless entropy target such as $H(X)$ would be appropriate only when the goal is exact reconstruction of a discrete source. Autoencoders are typically evaluated under a lossy distortion metric such as MSE, PSNR-derived distortion, or a task-specific reconstruction loss. Therefore, the rate-distortion target is more aligned with the actual reconstruction objective.

A.7 Ideal target rate

The ideal target rate is defined as

$$H_{\text{target}}^* = R_X(D_{\text{sat}}), \quad (60)$$

where D_{sat} is a selected saturation distortion. The value D_{sat} represents the quality level at which additional improvement is no longer considered practically important for the intended application.

The expression $H_{\text{target}}^* = R_X(D_{\text{sat}})$ is theoretically clean, but $R_X(D)$ is usually unknown for real datasets. It is therefore used as the ideal definition, while experiments use an operational approximation based on a reference compressor.

A.8 Reference compressor target

The practical target rate is

$$H_{\text{target}} = \hat{R}_{\text{ref}}(D_{\text{sat}}). \quad (61)$$

For a validation set $\{x_i\}_{i=1}^n$, a reference compressor operating near distortion D_{sat} gives

$$\hat{R}_{\text{ref}}(D_{\text{sat}}) = \frac{1}{n} \sum_{i=1}^n \text{bits}_{\text{ref}}(x_i; D_{\text{sat}}). \quad (62)$$

This is the average number of bits used by the reference compressor at the target distortion. It is not claimed to equal the unknown $R_X(D_{\text{sat}})$ exactly. Instead, it is an operational estimate of the rate needed by a concrete compression procedure.

A good reference compressor should be matched to the data type and distortion metric. If several references are available, the minimum measured rate among those that satisfy the distortion constraint can be used:

$$\hat{R}_{\text{ref}}^{\min}(D_{\text{sat}}) = \min_{m \in \mathcal{M}: \hat{D}_{\text{ref}}(m) \leq D_{\text{sat}}} \hat{R}_{\text{ref}}(m). \quad (63)$$

This reduces the risk that a single inefficient reference overestimates the target rate.

A.9 Saturation dimension

The proposed bottleneck dimension is

$$d_{\text{sat}} = \min \left\{ d \in \mathcal{D} : R_{\text{fac}}(d) \geq \hat{R}_{\text{ref}}(D_{\text{sat}}) \right\}. \quad (64)$$

For each candidate dimension d , the quantized latent rate $R_{\text{fac}}(d)$ is estimated on validation samples. The selected dimension is the smallest candidate whose measured latent rate reaches the reference rate.

This definition should be interpreted carefully. The dimension d_{sat} is not a proof of the smallest possible latent dimension. It is also not a guarantee of perfect reconstruction. It is a design-time estimate of the point at which the latent coding budget reaches an operational rate associated with the target distortion. The expected effect is that, beyond this point, additional dimensions tend to have diminishing reconstruction-quality gains.

A.10 Rate-distortion necessary condition

The quantized AE induces the Markov structure

$$X \rightarrow \hat{Z} \rightarrow \hat{X}, \quad (65)$$

where $\hat{X} = g_\phi(\hat{Z})$. If the AE achieves distortion at most D , then the reconstruction $\hat{X}$ is feasible for the rate-distortion problem at distortion level D . Hence,

$$R_X(D) \leq I(X; \hat{X}). \quad (66)$$

By data processing,

$$I(X; \hat{X}) \leq I(X; \hat{Z}). \quad (67)$$

Because $\hat{Z}$ is discrete,

$$I(X; \hat{Z}) \leq H(\hat{Z}). \quad (68)$$

Finally, by subadditivity of entropy,

$$H(\hat{Z}) \leq \sum_{j=1}^d H(\hat{Z}_j) = R_{\text{fac}}(d). \quad (69)$$

Combining the inequalities gives

$$R_X(D) \leq I(X; \hat{X}) \leq I(X; \hat{Z}) \leq H(\hat{Z}) \leq R_{\text{fac}}(d). \quad (70)$$

Therefore,

$$\mathbb{E}[\ell(X, \hat{X})] \leq D \implies R_{\text{fac}}(d) \geq R_X(D). \quad (71)$$

This is a necessary condition. It says that a quantized AE cannot achieve distortion D unless its factorized latent rate is at least as large as the rate-distortion lower bound. The converse is not guaranteed: having enough measured rate does not by itself ensure that the model class and optimizer use that rate efficiently.

A.11 Saturation interpretation

The final sizing rule can be summarized as

$$R_X(D_{\text{sat}}) \approx \hat{R}_{\text{ref}}(D_{\text{sat}}) \leq R_{\text{fac}}(d) = \sum_{j=1}^d H(\hat{Z}_j). \quad (72)$$

The left side represents the target rate required to reach a chosen reconstruction quality level. The right side represents the operational latent rate supplied by the quantized AE under the factorized coding protocol. The crossing point is used as the predicted saturation dimension.

The saturation claim is conditional. It is strongest when the target distortion lies near a flat region of the rate-distortion curve. If the distortion-rate curve changes slowly after a rate R_{sat} , then adding more rate produces only small distortion improvement. Under that condition, once $R_{\text{fac}}(d)$ reaches the reference target, additional dimensions are expected to produce diminishing returns.

A.12 What the equations do not claim

The formulation intentionally avoids three stronger claims. First, it does not claim that d_{sat} is the exact minimum possible autoencoder dimension. Second, it does not claim that $R_{\text{fac}}(d) \geq \hat{R}_{\text{ref}}(D_{\text{sat}})$ guarantees target distortion for every architecture and optimizer. Third, it does not claim that the factorized rate equals the true joint entropy when latent coordinates are dependent.

The intended claim is narrower and operational: under a fixed quantization, normalization, distortion metric, and factorized coding protocol, the crossing

$$R_{\text{fac}}(d) \geq \hat{R}_{\text{ref}}(D_{\text{sat}}) \tag{73}$$

predicts the onset of the reconstruction-quality saturation region.

Reproducibility Checklist

A submission based on this draft should report:

- all candidate dimensions $\mathcal{D}$;
- quantizer bin widths, clipping ranges, and normalization protocol;
- entropy estimator and validation sample size;
- reference compressor settings and measured rate-distortion points;
- random seeds and training budgets;
- whether models are independently trained or nested;
- total-correlation estimates or dependence diagnostics;
- confidence-aware entropy bounds when used.

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